Cross-Modal Attention for Dutch Grid Load Forecasting

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1 Introduction

Current grid load forecasting models for the Netherlands rely on historical load data, standard meteorological data [1, 4, 15], and economic indicators [1, 4, 15], failing to capture the high-dimensional socio-economic signals encoded in VIIRS *Nighttime Light* (NTL) imagery [2, 3, 10, 16]. Within the fields of energy informatics, sustainability, and remote sensing, research has emerged that validates the potential of *Multi-Modal Learning* (fusing heterogeneous data such as text, images, and time-series)—specifically incorporating VIIRS nighttime light imagery—in improving grid load forecast accuracy [2, 3, 10]. Furthermore, attention-based mechanisms have been tested as means of capturing complex inter-modal relationships to enhance time series forecasting for other domains, such as energy generation forecasting [12] or supply chain demand forecasting [16].

This study proposes to target the research gap by designing and testing cross-modal attention-based mechanisms (adapting architectures such as [12, 16]) that fuse NTL and time-series modalities for the Dutch context. These will be compared against established Dutch baselines (e.g., Prophet-LSTM [15], SARIMAX-LSTM [1]), utilizing ablation studies as in [16] to quantify the specific predictive gain attributed to the inclusion of VIIRS imagery and attention layers. Existing multi-modal architectures, such as SolarCrossFormer [12] and Causal-Aware Multimodal Transformer [16], shall serve as architectural references to be adapted for the Dutch grid load forecasting problem statement. Accessible public datasets (ENTSO-E, NASA Black Marble, KNMI) shall serve as the basis for the work; no new novel data collection is required.

Research Question: *How does the performance of a multi-modal attention-based deep learning model, integrating VIIRS nighttime light data, economic indicators, and weather data, compare to current unimodal (time-series only) baselines for electricity grid load forecasting in the Netherlands?*

To answer this main question, the following sub-questions are formulated:

- **SQ1 (Architectural Adaptation):** How can existing multi-modal attention architectures (e.g., SolarCrossFormer/CAMT) be adapted to fuse high-dimensional VIIRS satellite imagery with low-dimensional economic and weather time-series data for the specific context of the Dutch power grid?
- **SQ2 (Benchmarking):** How does the proposed multi-modal model compare in performance to established hybrid baselines for the Netherlands (specifically Prophet-LSTM and SARIMAX-LSTM) in terms of predictive accuracy (RMSE, MAPE) over short-to-medium term horizons?

Table 1: Comparison of Related Work and Research Gap

Study	Multi-modal (Img+TS)	Attention Mech.	VIIRS NTL	Econ & Weather	NL Context	Grid Load?
Wang [16]	✓	✓	✓	✓	×	×
Schubnel [12]	✓	✓	×	✓	×	×
Guo [6]	×	✓	×	✓	×	✓
Qiu [9]	×	✓	×	×	×	✓
R. et al. [10]	×	×	✓	×	×	✓
Chen [3]	×	×	✓	×	×	✓
Borunda [2]	×	×	✓	×	×	✓
v.d. Sande [15]	×	×	×	✓	✓	✓
Ashtar [1]	×	×	×	✓	✓	✓
Curiël [4]	×	×	×	✓	✓	✓

- **SQ3 (Ablation Analysis):** What is the specific predictive contribution of the VIIRS nighttime light imagery and the cross-modal attention mechanisms, as quantified through systematic ablation studies?

2 Related Work

Recent advancements in grid load forecasting within the Netherlands have predominantly focused on hybrid ensemble models. Studies by van de Sande et al. [15], Ashtar et al. [1], and Curiël et al. [4] have successfully demonstrated the efficacy of architectures such as Prophet-LSTM and SARIMAX-LSTM. These approaches rely on historical load data combined with standard meteorological data and traditional economic indicators. However, debate persists in the literature regarding the trade-off between the interpretability of these statistical hybrids versus the superior accuracy of "black-box" deep learning models [13]. The broader forecasting domain has begun shifting toward multi-modal learning. Research has explored the integration of diverse data sources, such as text and time-series for supply chain demand forecasts [16] or satellite imagery for solar irradiance forecasts [12]. Specifically, attention-based mechanisms have proven effective in capturing complex inter-mode relationships in these external domains [7]. Despite this, the utility of integrating high-latency exogenous variables, such as VIIRS nighttime light imagery, versus relying solely on endogenous load dynamics remains a point of contention [2].

Consequently, a gap exists in the current literature: there is a lack of models designed to simultaneously integrate VIIRS nighttime light imagery, economic indicators, and weather data via cross-modal attention-based mechanisms specifically for the Dutch context. As illustrated in Table 1, no existing study fulfills all these criteria. This research aims to bridge this gap by adapting multi-modal architectures similar to Wang et al. [16] and Schubnel et al. [12], while benchmarking results against established Dutch baselines [1, 15] using comparative metrics (RMSE, MAPE, MAE) and

ablation studies as in [16] to quantify the predictive gain of the visual and attention components.

3 Methodology

This research employs a quantitative, multi-modal research design to evaluate the predictive gain of integrating high-dimensional satellite imagery into Dutch grid load forecasting.

3.1 Data Procurement, Integration, and Preprocessing

This multi-modal forecasting architecture integrates four data streams: grid load, high-resolution satellite imagery, macroeconomic indicators, and local meteorology. All inputs are derived from validated, open-access repositories. Table 2 outlines the selected datasets. Distinct spatial and temporal granularities necessitate rigorous alignment prior to model input tensor formulation.

Table 2: Overview of Data Sources

Source	Description	Spatial Res.	Temp. Res.	Timespan
ENTSO-E	Grid load & net consumption	NL & regional zones	Hourly	2006–Present
NASA (VNP46A1)	Raw TOA Nighttime Lights (NTL)	15 arc-sec (~500m)	Daily	Jan 2012–Present
NASA (VNP46A2)	BRDF-adjusted, corrected NTL	15 arc-sec (~500m)	Daily	Jan 2012–Present
CBS	Energy prices, GDP, population	National (NL)	Monthly/ Yearly	1995–Present
KNMI	Meteorological observations	~50 stations	10-Min/ Hourly	1951–Present

Endogenous Target Variable: ENTSO-E Grid Load. The target variable is hourly electricity load in the Netherlands (2006–2025) [5]. This temporal resolution captures long-term macroeconomic shifts (e.g., COVID-19, electrification) as well as diurnal consumption cycles critical for day-ahead dispatch.

Exogenous Spatial Variable: VIIRS Nighttime Lights (NTL). We utilize the NASA Black Marble VNP46A2 product [8]. Its lunar BRDF correction prevents the model from conflating natural moonlight with anthropogenic activity [8]. NTL intensity and distribution serve as a high-dimensional proxy for regional economic vitality and industrial baseload demand [2, 3] (e.g., Port of Rotterdam).

Exogenous Temporal Variables: Socioeconomic Indicators (CBS). CBS data (consumer energy prices, GDP, population) establishes the structural baseline for energy consumption [14]. Including these variables allows the model to anticipate demand curtailment during extreme inflation or geopolitical instability.

Exogenous Temporal Variables: Meteorological Data (KNMI). Grid load fluctuates significantly with heating and cooling needs. We integrate high-fidelity in-situ observations (temperature, humidity, wind speed, precipitation) from over 50 automated KNMI weather stations across the Netherlands [11].

Aligning multi-frequency data without introducing systemic bias or leakage involves the following key steps, subject to expansion and refinement as the project progresses:

- **Temporal Up-sampling & Forward-Filling:** Lower-frequency socioeconomic data is up-sampled to hourly intervals using a strict forward-filling approach to prevent data leakage.
- **Image Patching & Spatial Aggregation:** To prevent Vision Transformer (ViT) overfitting and memory bottlenecks, VNP46A2 imagery is cropped to a Dutch bounding box and processed using fixed-size patch embeddings.
- **Outlier Mitigation & Imputation:** Missing data is imputed via Multivariate Imputation by Chained Equations (MICE). Sensor-induced load anomalies are capped at the 1st and 99th percentiles using IQR-based Winsorization to preserve data integrity.
- **Feature Scaling:** Continuous temporal features are Min-Max normalized to optimize gradient descent. Scaling parameters (X_{min}, X_{max}) are fit exclusively on the training fold to maintain strict separation from testing data.

3.2 Architectural Adaptation (SQ1)

To successfully fuse high-dimensional spatial imagery with low-dimensional time-series data, the proposed architecture discards naive early-fusion (concatenation) in favor of a dynamic Cross-Modal Attention Transformer (CMAT). The design is an adaptation of the *SolarCrossFormer* [12] and *Causal-Aware Multimodal Transformer (CAMT)* [16] frameworks, specifically engineered to process the disparate granularities of Dutch grid load, weather, and socio-economic indicators.

As illustrated in Figure 1, the architecture processes incoming data streams through three isolated, modality-specific encoders before fusion:

- **Time-Series Encoder:** Historical ENTSO-E grid load and KNMI meteorological observations are processed through a Temporal Transformer. By applying Multi-Head Self-Attention (MSA) along the temporal axis, this encoder captures the auto-regressive properties and periodic daily/weekly seasonalities of electricity demand independently of external spatial factors.
- **Visual Encoder:** Processing satellite imagery in deep learning poses a high risk of catastrophic overfitting. Following the empirical findings of Schubnel et al. [12], who demonstrated that full Vision Transformers (ViT) create significant computational overhead and tend to memorize pixel arrays rather than extrapolating broad spatial dynamics, this architecture omits a full ViT backbone [12]. Instead, the VIIRS NTL imagery undergoes a Linear Patch Embedding process [12]. The geographic bounding box is divided into non-overlapping patches and linearly projected into a latent dimension matching the time-series encodings, preserving localized node data without overwhelming the network.

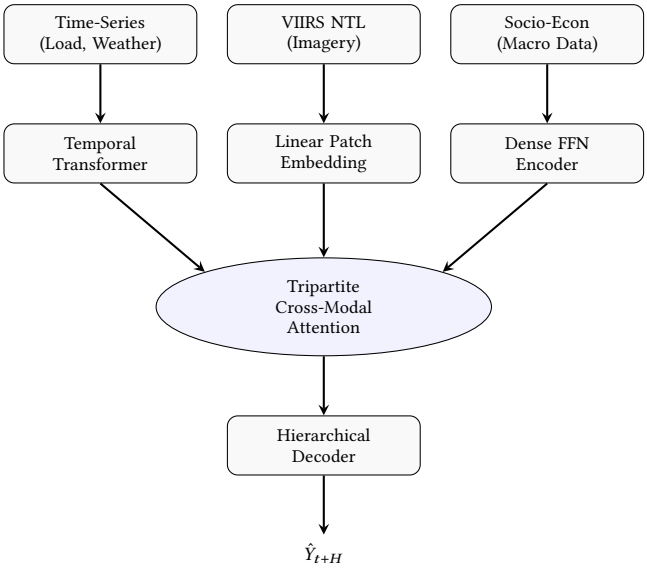

Figure 1: Proposed Cross-Modal Attention Transformer (CMAT) architecture, featuring modality-specific encoders and tripartite attention. Adapted from [12, 16].

- **Socioeconomic Encoder:** Low-frequency, low-variance macroeconomic indicators (such as CBS price indices and GDP) are projected into the shared latent space via a series of dense Feed-Forward Networks (FFN).

Following independent encoding, the representations are integrated via a **Tripartite Cross-Modal Attention** mechanism [16]. Traditional transformers utilize self-attention, mapping a sequence strictly against itself. In contrast, this cross-attention layer allows the queries (Q) derived from the temporal weather and load encodings to attend dynamically to the keys (K) and values (V) extracted from the visual satellite patches and economic data, and vice-versa. This mechanism adaptively weights the modalities based on the prevailing context as in [16]; for example, the network may lean heavily on the visual encoder's representation of industrial nighttime lighting during periods of stable weather, but shift attention entirely to the time-series encoder during severe winter temperature drops.

Finally, the fused cross-modal latent representation is passed through a hierarchical decoder to output the point load forecasts $\hat{Y}_{t+H}$ across the defined prediction horizon.

3.3 Benchmarking and Evaluation (SQ2)

To isolate the value of VIIRS satellite imagery and cross-modal attention, our architecture is benchmarked against state-of-the-art unimodal models from recent Dutch grid forecasting literature. These baselines are trained on identical temporal, meteorological, and economic data, completely omitting the VIIRS visual modality:

- **Prophet-LSTM:** Decouples seasonal/trend components via Prophet, feeding residuals into an LSTM to capture nonlinearities. Serves as the primary unimodal benchmark, having historically achieved high correlation ($r = 0.93$) for Dutch winter demand.
- **SARIMAX-LSTM:** Combines classical statistical transparency for linear trends with deep learning for nonlinear error correction. Previously achieved a 1.88% Mean Absolute Percentage Error (MAPE) at a 180-day horizon.
- **Stacked Prophet-QLSTM:** Represents the cutting edge of classical ensemble forecasting, utilizing XGBoost-based stacked generalization to aggregate predictions.

Because standard k-fold cross-validation induces severe data leakage in time-series data, we employ Rolling-Origin Cross-Validation (ROCV). Point forecasts are evaluated using a multi-metric approach as in literature [1, 4, 15] to capture different facets of predictive error:

- Root Mean Square Error (RMSE): Heavily penalizes large prediction errors (which could trigger expensive emergency reserves). Minimizing RMSE is a critical operational objective.
- Mean Absolute Percentage Error (MAPE): Provides an intuitive, scale-independent metric.
- Mean Absolute Error (MAE): Applies a linear penalty to errors, offering a robust indicator of average day-to-day forecast reliability that is less skewed by extreme outliers.
- Pearson Correlation Coefficient (PCC, r): Measures linear correlation between the predicted and actual load trajectories. It evaluates the model's skill in capturing rhythmic patterns (peaks, troughs) independent of absolute scalar magnitude bias.

3.4 Ablation Analysis (SQ3)

To isolate the contribution of each component (SQ3), we conduct a systematic ablation study as in [16]. By incrementally adding or removing architectural components and data modalities, the marginal utility of each innovation can be mathematically isolated and evaluated against the RMSE and MAPE metrics. The following distinct model configurations will be trained and evaluated under the identical ROCV protocol:

(1) **TS Only:** Unimodal baseline (Load + Weather).
(2) **TS + Econ:** Bimodal configuration excluding imagery.
(3) **TS + Visual:** Bimodal configuration excluding economic data.
(4) **Early Fusion:** All data included via naive concatenation (Attention disabled).
(5) **Full CMAT:** The proposed model with cross-modal attention.

Performance differences between C3 and C1 will quantify the marginal utility of VIIRS NTL, while the C5–C4 delta will validate the efficacy of the attention-based fusion mechanism.

3.5 Optional Extension: Causal discovery

To enhance interpretability, we propose a secondary **Causal Discovery Module** based on the CAMT framework [16]. This module

applies Granger Causality principles via a sparsity-penalized network to learn a directed acyclic graph (DAG) of the features [16]. This allows the model to justify a predicted load change by identifying its primary causal driver (e.g., a specific socio-economic signal or weather event), moving beyond purely associative "black-box" forecasting.

4 Risk Assessment

While no new primary data collection is required, integrating heterogeneous multi-modal datasets presents potential technical and scheduling risks:

- **Data Alignment and Memory Bottlenecks:** Processing high-resolution spatial imagery (VIIRS) alongside time-series data can quickly exceed available GPU memory and significantly slow down training epochs. *Mitigation:* We will strictly enforce spatial cropping to the Dutch bounding box and utilize linear patch embeddings instead of a full Vision Transformer (ViT). Access to institutional compute clusters will be secured as soon as possible.
- **Architecture Convergence Issues:** Fusing disparate modalities via cross-modal attention is mathematically complex; the model may initially struggle to converge or fail to outperform established statistical baselines. *Mitigation:* We will adopt an iterative development approach. Simple early-fusion (concatenation) will be established first as a fallback. Furthermore, the ablation study design ensures that even a null or negative performance result regarding the visual modality provides a rigorous academic contribution.
- **Scope Creep:** The integration of the optional Causal Discovery module may threaten the core timeline. *Mitigation:* This module is strictly designated as a secondary extension. If the primary Cross-Modal Attention Transformer (CMAT) experiments are not completed by appropriate milestones, causal discovery will be excluded from the final thesis to ensure on-time submission.

5 Project Plan

Given that the Thesis Design phase is concluding in late February, the project is structured across the remaining 17-week timeline from March to the final deadline in late June. The chronological pipeline aligns technical execution—from data preprocessing to hyperparameter tuning—with the strict institutional milestones required by the University of Amsterdam.

Critical buffers are integrated into June to account for formatting, revisions, and deep learning training overheads. The detailed timeline is presented in the Gantt chart (Figure 2).

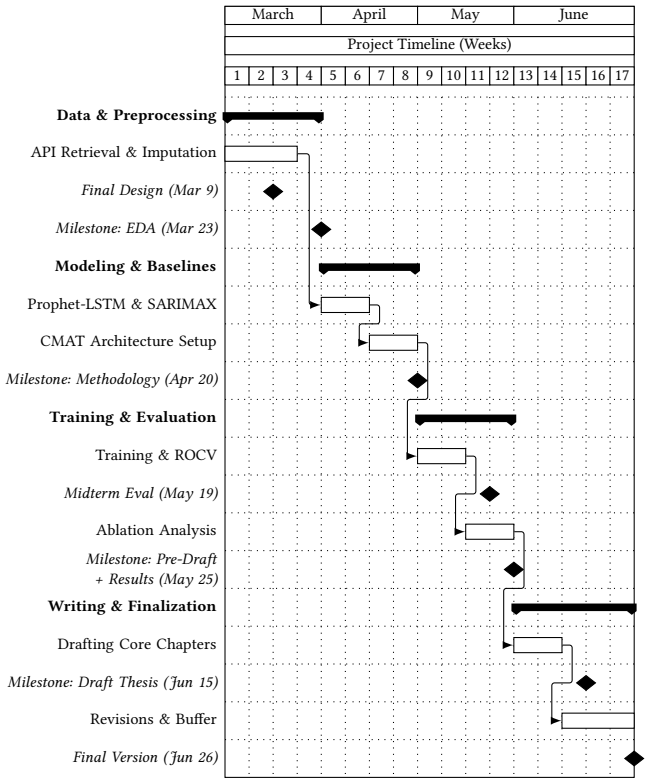

Figure 2: 17-Week Project Timeline tracking technical implementation against official UvA milestones (March–June).

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